

AI & THE FUTURE OF WORK

WHAT KEEPS MANY UP AT NIGHT LATELY?

- Will there be job opportunities for me when I graduate?
- Will I be laid off, with my role *replaced by AI*?

THE ANSWER

- You will be replaced by someone who can **use AI better than you**

DS VS AI

- ***Data science (DS)*** and ***machine learning (ML)*** provide key foundations for ***artificial intelligence (AI)***
 - ML is a core subfield of AI and underpins most modern AI systems.
 - DS supplies the data handling, statistical analysis, and modeling practices that make ML and AI practical and effective.

MODULE DELIVERY: TRADITIONAL APPROACH

- Traditional DS modules provide systematic and **comprehensive coverage** of a programming language (such as Python or R)
 - building knowledge firmly from the ground up
- ***Downside:***
 - Buried into details, students can easily **lose the big picture**,
 - forgetting why it is worth their effort to learn a DS programming language

OUR STRATEGY

- Given the breath of knowledge this module will expose you to,
 - ***one term is too short*** for developing the knowledge comprehensively in the traditional manner
- Instead, this module simply showcases minimal examples of the latest best-in-class DS and AI tools
 - ***Best-in-class***: The core concepts underpinning these tools will last
 - ***Minimal examples***: Students can appreciate why the tools are the best in class without losing the big picture

THE GOAL

- Is NOT to turn students into proficient coders
- Rather, the module aims to:
 - Inspire students to independently explore selected tools, guided by their **own interests and motivation**
 - Signpost resources that support **self-directed exploration** by students

DELIVERY FORMAT

- Interactive, *hands-on workshop* sessions
 - Week 1-4, 6-10
- Revision session (*recap of key takeaways*)
 - Week 11 (Thursday session)
 - Week 12 (online, replacing the Easter Monday sessions)
- ***Optional*** drop-in helpdesk sessions
 - Weeks 6–11
 - No formal teaching
 - For clarification questions only

EXPECTATION ON GROUP ASSIGNMENT

- Students must demonstrate competence in the following skills to achieve marks in each range:
 - **40-54:** Coding and data skills covered in Week 1 to 4
 - **55-69:** Not only the above but also
 - **basic** additional skills from self-exploration of closely related tools and techniques for any week(s)
 - **70+:** Not only the above but also
 - **enhanced** additional skills from self-exploration of closely related tools and techniques for any week(s)

Note: Within each mark range, students who more convincingly demonstrate higher competence in their **written report (approx. 1,500 words)** will earn marks closer to the upper end of the range.

EXPECTATION ON INDIVIDUAL ASSIGNMENT

- Students must demonstrate competence in the following skills to achieve marks in each range:
 - **40-54:** Coding and data skills covered in Week 1 to 4
 - **55-69:** Not only the above but also
 - i. additional skills from self-exploration of closely related tools and techniques for any week(s), **OR**
 - ii. skills covered in Week 6 and 10, **OR**
 - iii. both (i) and (ii)
 - **70+:** Not only the above but also
 - skills covered in Week 7 to 9 **AND** additional skills from self-exploration of closely related tools and techniques for any week(s)

Note: Within each mark range, students who more convincingly demonstrate higher competence in their **video report (max. 3 minutes)** will earn marks closer to the upper end of the range.

OVERVIEW OF THE TOPICS

WEEK 1 MODULE OVERVIEW AND PYTHON BASICS

Theme: *Building the Foundation*

- Why this topic
 - **Jupyter notebooks** provide an interactive environment mixing code, explanations, and outputs
 - suitable for beginners to experiment with Python basics
 - Introduce a modern **cloud-based coding environment** for data professionals
 - enhanced by AI coding assistants
 - Introduce basic concepts (e.g., data types, control flow statements) in a step-by-step manner,
 - laying the foundation for building more on it in future sessions

WEEK 1 CONT'D

- What you will learn
 - Navigate **GitHub Codespaces** and the Jupyter Notebook interface
 - Understand core Python data types (`int`, `str`, `list`, `bool`, `float`) and how to manipulate them for simple tasks
 - Use **conditionals** and **for-loops** to automate repetitive tasks

WEEK 2 AUTOMATED FINANCIAL ANALYSIS

Theme: *From Static Code to Dynamic Tools*

- Why this topic
 - To transition from manual, static calculations to **automated** computations for reproducibility and flexibility
 - **Marimo** builds on Jupyter with “reactivity” (i.e., changes automatically update outputs, like in Excel)
 - making it a best-in-class tool for iterative exploration without Jupyter’s limitations

WEEK 2 CONT'D

- What you will learn
 - Define and use functions to compute **Altman Z-Score** with fetched data and user inputs
 - Z-Score is a credit risk measure
 - Use **visualizations** to facilitate Z-Score interpretation

WEEK 3 PANEL DATA AND INTERACTIVE PLOTS

Theme: *Handling Complexity of Panel Data & Interactive Visuals*

- Why this topic
 - Real-world financial analysis often goes beyond a single company or point in time, requiring the use of
 - Cross-sectional, time-series, or **panel** (multiple companies over multiple years) data
 - Gathering such data manually is impossible at scale
 - Python **automates** the retrieval and organization of thousands of financial records

WEEK 3 CONT'D

- Why this topic (cont'd)
 - **Plotly** enables interactive, web-ready visualizations that go beyond static charts
 - especially useful for presenting insights engagingly
 - making abstract concepts tangible and shareable
- What you will learn
 - **Programmatically fetch** fundamental and stock data from Yahoo Finance
 - Create professional, **interactive** graphics (including 3D plots) to visualize financial metrics,
 - highlighting differences across years and sectors

WEEK 4 DATA PREPARATION AND MY FIRST WEB APP

Theme: *Turning Raw Data into Shareable Web Apps*

- Why this topic
 - Showcase how code can be turned into **deployable dashboards/apps**
 - bridging coding to practical career skills
 - Illustrate **data preparation** as a foundational step for deriving insights from data analysis

WEEK 4 CONT'D

- What you will learn
 - Clean and transform data (e.g., handle missing values, construct new variables)
 - Build simple web outputs to display prepared data and visualizations
 - Assemble outputs into a **multi-tabbed webpage** featuring an interactive chart and dashboard
 - **Publish** the dynamic webpage to **GitHub Pages** so anyone can view it online

WEEK 5 READING WEEK

- No class

WEEK 6 DATA ANALYSIS WORKFLOW

Theme: *From Raw Data to Rigorous Insight*

- Why this topic
 - Walk through a *data analysis workflow*:
 - from data collection, preparation (like constructing Tobin's q and leverage) to **regression analysis**,
 - illustrating how data analysis is done in academia and industry

WEEK 6 CONT'D

- What you will learn
 - Access and merge professional financial data (**Compustat**) with ESG metrics
 - **Profile data** to identify missing values and outliers
 - Run a regression analysis to answer the question:
 - *Does superior ESG performance increase Firm Value (Tobin's q)?*

WEEK 7 GATHERING UNSTRUCTURED DATA

Theme: *Mining the Web for Hidden Signals*

- Why this topic
 - Unstructured data, like company sustainability reports in PDF format, contains rich insights (e.g., on environmental topics like water usage)
 - but is hard to collect due to web challenges like **bot detection** and **cookies**
 - **Automating collection** with Python tools enables efficient handling of large-scale unstructured data
 - a key skill for real-world applications in finance where manual processes are tedious and error-prone

WEEK 7 CONT'D

- What you will learn
 - Bypass bot detection and cookies
 - **Web crawl** to locate report URLs
 - Download PDF reports programmatically
 - **Extract pages** containing specific keywords (e.g., “water”)

WEEK 8 INTRO TO LARGE LANGUAGE MODELS

Theme: *AI as Your Analytical Co-Pilot*

- Why this topic
 - LLMs represent cutting-edge AI for natural language tasks,
 - like extracting and classifying **sentiment** in forward-looking statements from annual reports to reveal firm outlooks on growth, risks, and performance-key signals for investors and analysts
 - Understanding basics demystifies AI, showing how it augments (not replaces) human work by **automating text analysis** while requiring critical evaluation for accuracy in finance contexts

WEEK 8 CONT'D

- What you will learn
 - Interact with LLMs (via APIs like Groq) for tasks like classifying forward-looking sentences as optimistic, neutral, or cautious
 - Evaluate outputs critically, focusing on **prompts** for accurate financial queries, and experiment with parameters (e.g., **temperature** and **top_p**) to observe effects on reproducibility and creativity

WEEK 9 API ACCESS TO LLMS

Theme: *Automating Intelligence at Scale*

- Why this topic
 - Application Programming Interface (API) enables **programmatic access to LLMs**, integrating AI into custom tools (e.g., chaining prompts for refined sentiment scoring)
 - a **scalable**, enduring skill for efficiency in automating financial text processing.
 - Build on intro concepts, showing real integration for tasks like optimizing classifications to better inform investment recommendations based on firm tones

WEEK 9 CONT'D

- What you will learn
 - Set up API calls to LLMs (e.g., Groq or OpenAI) for structured responses, incorporating **few-shot** examples (positive/negative/neutral sentences) and chaining for multi-step analysis
 - **Chain** API outputs with Python for enhanced workflows, like using DSPy to programmatically optimize prompts and an “AI judge” to validate sentiment classifications

WEEK 10 ADDITIONAL TOOLS & RESOURCES

Theme: *Expanding Your Digital Toolkit*

- Why this topic
 - No single tool solves every problem
 - Understanding the “right tool for the job” is a key skill for data professionals
 - The landscape of Data Science and AI **changes rapidly**
 - Knowing where to find resources is essential for continuous **self-learning and exploration**

WEEK 10 CONT'D

- What you will learn about
 - **Polars:** Next-generation data processing for high-performance and “big data” tasks (beyond Pandas)
 - **R Language:** The industry standard for specialized statistical analysis and econometrics (vs. Python)
 - The Batch newsletter and courses on **DeepLearning.AI**

WEEK 10 CONT'D

- What you will learn about (cont'd)
 - Other cloud-based coding platforms
 - **RStudio on Posit Cloud:** For best-in-class R support and one-click deployment of interactive dashboards, reports, and APIs
 - **Google Colab:** For free, limited access to TPUs & GPUs
 - **Lightning.ai:** For compute scalability and a unified ecosystem for AI fine-tuning and deployment (with monthly allowance for free GPU access)

FINAL THOUGHT

- You will leave with the **data literacy** to
 - **adapt to** any new tool **the future** of finance and accounting throws at you

Give yourself the best chance to
stand out in the talent pool!

AFTER THE BREAK

- Let's open the Jupyter notebook
`Wk01_Jupyter_PythonBasics.ipynb`
 - by double-clicking it in the Explorer panel
(on the left side of a Codespace)

GITHUB CODESPACES: LAYOUT

- Panel toggle buttons (upper right corner)
 - Buttons on Left Panel: **Explorer**, ..., Extensions, ...
 - Extensions button:
 - Make sure the **Jupyter extension** is installed and enabled
 - Tabs in Bottom Panel:
 - **Terminal** | ... | Jupyter (visible when a Jupyter notebook is opened) | ...
 - Right Panel: **AI chat box** with two drop-down menus
 - Mode selection (e.g., Ask, Edit)
 - To be conservative, “Ask” is recommended
 - Model selection (e.g., GPT-5 mini, Grok Code Fast 1)
 - Center Panel: **Code editor** area

JUPYTER NOTEBOOK: STRUCTURE

- Code Cells vs. Markdown Cells
 - **Code Cells:** Contain Python code to be executed
 - **Markdown Cells:** Contain formatted text, headings, ordered / unordered lists, links, etc.
- Select **Kernel** button (top right of center panel)
 - Must choose a Python 3.xx kernel before running code cells
- **Execution** button ()
 - Visible when hovering over/selecting a code cell
 - Click it to run the code in a code cell

WHAT HAVE YOU LEARNED?

- See this week's **Summary of Topics** document for a summary of the *Key Concepts, Coding Techniques, and Packages Introduced* covered in this session
- Your follow-ups
 - See this week's **Activities after the session** listed on Moodle
 - In particular,
 - go through items on the *Read/Watch After* list of this week's **Read/Watch in Advance and After** document